ARNAUD GERMAIN

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Personal information
Birth year: 1999
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ACADEMIC POSITION

NEOMA Business School, Assistant Professor of Finance *Aug. 2026–*

RESEARCH INTERESTS

Machine learning in finance, credit risk, portfolio selection

EDUCATION

UCLouvain, PhD in Economics and Management Sciences *Dec. 2022–July 2026*
Thesis title: *On Clustering for Credit Risk Management: Applications to Loan Selection and Default Prediction*
Supervisor: Frédéric Vrins (UCLouvain)
Jury: Raffaella Calabrese (University of Edinburgh), Paolo Giudici (University of Pavia), Corentin Vande Kerckhove (UCLouvain), Bertrand Candelon (UCLouvain)

KU Leuven, M.Sc. in Business Engineering* – Data Science, *summa cum laude* *2020–2022*

UCLouvain, M.Sc. in Business Engineering* – Financial Engineering, *magna cum laude* *2020–2022*

*As part of the Double Degree program between UCLouvain and KU Leuven.

UCLouvain, B.Sc. in Business Engineering, *cum laude* *2017–2020*

PUBLICATIONS

Credit Selection in Collateralized Loan Obligations: Efficient Approximation Through Linearization and Clustering [Link]
Co-author: Frédéric Vrins
European Journal of Operational Research – ABS 4

Despite its role in the global financial crisis, collateralized loan obligation (CLO) remains a powerful tool to direct funds towards the real economy. In particular, it enables development banks to increase credit supply to SMEs. Public financial institutions thus face the challenge of identifying a subset of credits to be pooled in a CLO for the sake of reaching a specific financial target. This is a mixed-integer nonlinear program, known to be NP-hard. In this paper, we provide an efficient method to tackle this problem by relying on the large pool approximation combined with clustering and linearization of ancillary variables. As illustration, we consider two objective functions. We rely on the celebrated one-factor Gaussian copula in the main examples, but make clear that this assumption is not a restriction and can be relaxed. Our results contribute to reducing the funding cost of SMEs and are of direct interest to securitization stakeholders such as public financial institutions, commercial banks, and pension funds.

WORK IN PROGRESS

Early Warning System for Non-Performing Clients [Link]
Co-author: Frédéric Vrins

In its "Guidance to banks on non-performing loans," the ECB requires banks to implement an Early Warning System (EWS) to identify potential non-performing clients at a very early stage. Relying on a unique dataset provided by a systemic European bank, including 5.5 million observations of anonymized data from 2018 to 2022, we aim to predict the corporate clients who will become non-performing within a given warning horizon. We propose two solutions to address time and client heterogeneity. First, we divide our dataset into several clusters using k-means, fit a prediction model on each cluster, and combine those models. This boosts out-of-sample performance relative to fitting a single prediction model on the full dataset or relying on domain knowledge to determine the clusters. Second, we forecast the unconditional probability of a positive outcome using macroeconomic variables and rescale the prediction model's output using Bayes' theorem. This improves out-of-sample performance relative to directly including the macroeconomic variables as predictors. The two approaches are complementary, and their combination yields the best predictive performance. Our findings help improve the performance and robustness of EWS.

Clagging: Generating and Combining Predictions Using Clustering [Link]

We consider an ensemble learning strategy called *Cluster aggregating (Clagging)*, where we (i) cluster the training set, (ii) fit a model on each cluster, and (iii) combine the predictions of each model. We introduce a new combination scheme exploiting the distance of a test point to the clusters' centroids. The intuition behind using clustering rather than random bootstrapping or partitioning is to create *meaningful diversity*. We conduct an extensive horse-race study benchmarking Clagging against state-of-the-art methods on 20 datasets. Our empirical evidence suggests that Clagging often outperforms bagging, where bootstrapped samples are traditionally created by drawing observations with replacement until their size matches that of the original training set.

SCHOLARSHIPS AND PRIZES

ING Chair 2022–2026

Four-year fully funded scholarship for a PhD thesis.

Rotary Chimay-Couvin Rhetorical Contest 2017

Audience award, *Tournoi des idées*.

Rallye Math 2016–2017

Mathematics contest organized by Université de Namur – 1^{st} *place both years*.

CONFERENCE TALKS

2026	17th Actuarial and Financial Mathematics Conference – AFMath (Brussels); European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (Bruges).
2025	41st International Conference of the French Finance Association (University of Dijon); 14th International Conference of the Financial Engineering and Banking Society (MBS); Credit Scoring and Credit Control Conference XIX (University of Edinburgh).
2024	International Conference on Computational Finance – ICCF (University of Amsterdam); 15th Actuarial and Financial Mathematics Conference – AFMath (Brussels).

TEACHING EXPERIENCE

NEOMA Business School *2026–2027*

Evaluation et mesure de performances financières; *Financial Data Analytics & Programming*; *Finance: Investing and Trading.*

UCLouvain *2023–2026*

Teaching Assistant, *Credit and Interest Rate Risk* (graduate, English).

WORK EXPERIENCE

Euroclear Intern in capital modelling: summer 2021.